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Conformational changes of cyclohexanethiol adsorbed on gold surfaces

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Abstract

The conformational changes of cyclohexanethiol (CHT) on gold nanoparticles and gold plate surfaces were investigated by means of surface-enhanced Raman scattering. The equatorial chair conformer appeared to be dominant at high bulk concentrations, whereas, both the equatorial and axial chair conformers of CHT were found to exist at low bulk concentrations corresponding to a near monolayer coverage limit. Molecular-scale scanning tunneling microscopy imaging revealed the existence of two axial and equatorial conformers in the CHT monolayer on an Au(1 1 1) surface as well as the existence of the meta-stable conformer, which may be due to dynamic conformational changes of the cyclic ring. It was also found that the peaks corresponding to the axial bands disappeared below room temperature, indicating the equatorial conformer is dominant at low temperatures. From this result, we found that the conformation of CHT on an Au surface was easily changed by surface temperature.
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1. Introduction

Conformational changes of molecules adsorbed on surfaces have been extensively studied to gain an understanding of their structure and dynamics which are of importance in the areas of biology, catalysis, material science and surface science [1]. The dependence of energy states on the physical conditions of various conformers should also provide valuable information pertaining to the locally stabilized structure [2]. Adsorbates on metal surfaces have been found to show conformational switching when investigated by means of spectroscopic tools [3].

Surface-enhanced Raman scattering (SERS) is one of the most sensitive techniques for monitoring adsorbates on metal substrates at the sub-monolayer coverage limit despite the equivocal selection rules and the fact that this application is limited to only a few metals (Ag, Au, and Cu) [4]. Electromagnetic (EM) surface selection rules provide a quantitative understanding of the interfacial structures of aromatic adsorbates on silver and gold nanoparticles [5]. Advances in scanning tunneling microscopy (STM) technologies have led to a variety of opportunities to explore, probe, and manipulate interactions at surfaces and interfaces [6]. A number of STM studies have revealed adsorption structures and phase transitions of self-assembled monolayers on various substrates at the molecular level [7,8].

Although colloidal dispersions of gold have been studied intensely since they were originally reported by Faraday [9], their physicochemical characterization has thus far not been performed systematically. Optical properties of gold particle aggregates have been extensively investigated by surface plasmon resonance, and their cluster sizes have been estimated using transmission electron microscopy (TEM) [10]. The particle diameter was controlled in the range of 9–99 nm by varying the amount of the citrate added for chemically reduced gold nanoparticles [11].

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Numerous cases have been reported indicating that the adsorption behavior and molecular packing arrangement depends strongly on the substrates [12–15]. However, the detailed origins of the different adsorption characteristics on Ag and Au have not yet been fully understood. One of the main purposes of our recent Raman studies is to understand how the adsorption behavior of various organic molecules changes with different metal substrates [5,14].

Cyclohexanethiol (CHT) is known to have various conformations from the chair-skew-boat and equatorial–axial equilibria of its ring to the different possible orientations of the thiol group [16]. By theory, the chair conformer is calculated to be more stable than the skew-boat conformer by 23 kJ mol^{-1}. Among the chair conformers, the equatorial conformation was predicted to be more stable than the axial conformation by 5 kJ mol^{-1}. A previous SERS study suggests that the conformations of CHT molecules on a sliver surface change from axial to equatorial with increasing surface coverage [17]. Recently, our STM study revealed that the adsorption of CHT molecules on Au(1 1 1) surfaces leads to well-ordered self-assembled monolayers (SAMs) even though CHT contains a high-flexible cyclic ring [18].

Although the adsorption and structural properties of CHT on silver characterized by SERS were previously reported [17], the conformational change of CHT on a gold surface has not been reported to date. The main purpose of this work is to elucidate the adsorption behavior and conformational dynamics of CHT molecules on gold nanoparticles and plates and their dependence on bulk concentration and surface temperature by means of SERS. In addition, the surface structure of a CHT monolayer on a Au(1 1 1) surface was examined by STM to understand the molecular-scale features of CHT in the monolayer.

2. Experimental

CHT (>98%) was purchased from Tokyo Kasei and used without further purification. Colloidal nanoparticles were prepared according to the procedures reported in the literature, wherein sodium citrate was used as a reducing agent [19]. All chemicals used were reagent-grade unless otherwise specified. Triply-distilled water of resistivity greater than 18.0 MΩ cm was used in making aqueous solutions. A 0.5 M ethanolic solution of CHT was added drop wise to 0.1–2 ml of a Au nanoparticle solution to a final concentration of 1×10^{-5}–9×10^{-5} M using a micropipet. The gold nanoparticles, initially purple, became bluish-green upon the addition of CHT. After obtaining a low concentration SERS spectrum of 1×10^{-5} M, cyclohexanethiol was subsequently added to a final concentration of 9×10^{-5} M. The Au substrate was prepared by the oxidation–reduction cycle. Before immersing the Au substrate in the ethanolic solution of CHT, the substrates were rinsed with excess ethanol and dried in an N_2 gas stream. After the self-assembly of the CHT monolayer, the substrate coated with the SAMs was rinsed with excess ethanol and dried in an N_2 gas stream again. For STM measurements, the atomically-flat Au(1 1 1) substrates on mica were prepared by vacuum deposition as reported in a previous paper [18]. The SAMs on Au(1 1 1)/mica were prepared by dipping the gold substrates in a 1 mM ethanol solution of CHT at room temperature for one day.

Raman spectra were obtained using a Renishaw Raman confocal system Model 1000 spectrometer equipped with an integral microscope (Leica DM LM). A 35 mW air-cooled He/Ne laser (Melles Griot Model 25 LHP 928) with a plasma line rejection filter provided the 632.8 nm irradiation excitation source used for the Au SERS experiments. Temperature-dependent Raman measurements were performed using a Linkam THMS 600 thermal heating stage. All STM images were acquired using a Nanoscope E (Veeco, CA) under constant current mode with Pt/Ir (80/20) at room temperature in air. All *ab initio* molecular orbital calculations were carried out using the Gaussian ′03 package. Geometry optimization of four CHT conformers was carried out at the level of B3LYP/6-31G++(d,p). The geometry optimization for the conformers was carried out starting from many possible orientations and no geometric constraints were used for the present calculations.

3. Results and discussion

Fig. 1a shows the ordinary Raman (OR) spectrum of CHT in the liquid state. The SERS spectrum for CHT molecules adsorbed on Au nanoparticles or Au plates is different from and more complex than the OR spectrum, as shown in Fig. 1b and c. We here note that since we could not find the vibrational bands for the axial conformer on Au plate at room temperature (see Fig. 6d), the spectrum

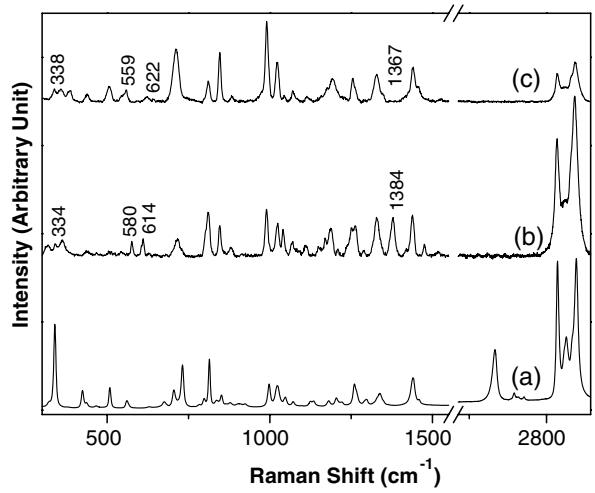

Fig. 1. (a) Ordinary Raman spectrum of CHT in the neat state. (b) SERS spectrum of 1×10^{-5} M CHT in aqueous gold nanoparticles and (c) on Au plate surfaces. Since we could not find the vibrational bands for the axial conformer on Au plate at room temperature, the spectrum obtained at 40 °C is presented for a better comparison. The spectral region between 1600 and 2400 cm^{-1} was omitted because it lacked relevant information. The frequencies of the most important diagnostic bands for the axial conformer are placed in the figure.

obtained at 40 °C is presented for a better comparison (Fig. 1c). The SERS spectrum of CHT adsorbed on Au nanoparticles (Fig. 1b) at room temperature is markedly different from that on Au plates (Fig. 6d). From this result, it is clear that conformation of CHT is strongly dependant on gold substrates, which may be resulting from the different morphology and roughness of gold surface. The main difference between the OR and SERS spectra is that the peak assigned to the S–H stretching band at 2570 cm^{-1} in the OR spectrum is absent in both the Au SERS spectra. Although not shown here, it was also found that the SERS spectrum on the Ag plate is analogous to that on Ag sol surfaces reported previously [17]. Since the SERS spectra on Ag and Au are found to be slightly different from each other presumably due to their dissimilar orientation on the surfaces, we compared the Au substrates with the same characteristic bands for axial conformer of ~340, ~570, ~620, and ~1380 cm^{-1} as placed in Fig. 1. It is also possible that CHT may have a slightly different orientation on Au nanoparticle and plate surfaces.

Together with the disappearance of the β(CSH) bending mode at 908 cm^{-1}, it is suggested that CHT adsorbs via the rupture of the S–H bonds on the Au surfaces, resulting in the formation of S–Au bonds. We analyzed the SERS spectrum referring to earlier vibrational assignments described in the literature [17,20] and the results are summarized in Table 1.

As previously discussed, a total of eight different conformers (equatorial and axial chair conformations and six skew-boat conformations) are possible when CHT adsorbs as a thiolate on surfaces. The chair conformers should be more prevalent than the skew-boat conformers as a result of an energy difference between them of 23 kJ mol^{-1}. While the equatorial chair conformer is expected to be more stable than the axial chair, their energy difference is only 5 kJ mol^{-1}; it is therefore likely that the two conformers coexist on the surface and that both affect the spectral features. Fig. 2 shows structural formula of two energetically stable conformers, i.e., equatorial and axial conformers, of CHT on a gold surface. Results of DFT calculations regarding the stabilities of the equatorial and axial forms of chair and boat conformers of CHT are summarized in Table 2. Our DFT calculation results show good agreement with those previously reported [16]. Although the overall spectral pattern of the SERS spectrum on Au nanoparticles appeared to be analogous to that on Ag nanoparticles, several vibrational band intensities showed significant differences. The vibrational bands at ~810, ~1040, ~1440, ~2850, ~2950 cm^{-1} exhibited much stronger intensity on Au than on Ag. Considering that all these vibrational bands could be assigned to the equatorial chair conformation, it is not certain whether such differences are due to different conformational ratios on Ag and Au. In Fig. 1b, several bands were ascribed to the axial chair conformer and were carefully verified by varying the bulk concentrations.

Previously it was reported that the intensities of the equatorial and axial chair conformers on a Ag surface depend on the bulk concentrations [17]. To examine the concentration-dependent conformational change, we obtained the SERS spectra for CHT adsorbed on Au nanoparticles at various concentrations. Fig. 3 shows the SERS spectra of CHT in the concentration range from 1×10^{-5} to 9×10^{-5} M. Several spectral changes were observed by varying the bulk concentration of CHT. The intensities of the vibrational bands at 344, 580, 614, and 1380 cm^{-1} assigned to the axial conformer weakened with increasing bulk concentrations. We found that both the equatorial and axial conformers coexist on Au surfaces in all concentration ranges. However, the amount of axial conformer showed a marked decrease in the range of 5–9×10^{-5} M. This suggests that the equatorial conformer

Table 1
Spectral data and vibrational assignments for CHT

Ordinary Raman	Au SERS		Tentative assignment[a]
	(~10^{-5} M)	(~9×10^{-5} M)	
(Blank)	238	238	ν(Au–S)
327	322	319	Eq. ring def.
(Blank)	344		Ax. ring def.
343	366	365	Eq. C–S bending
429			Eq. ring def.
440	440	438	Eq. ring def.
512	510	510	Eq. ring def.
565	580		Ax. ring def.
634	614		Skew. C–S str.[b]
679			Ax. C–S str.
709	(Blank)	(Blank)	Eq. C–S str.[c]
736	720	721	Eq. C–S str.[c]
788			Eq. C–H bending
802			Ax. ring breath
818	815	815	Eq. ring breath
840			C–SH bending
856	850	850	Eq. ring str.
860			Ax. ring str.
883	885	887	Eq. ring str.
908			C–SH bending
928			Ax. CH_2 rocking
1002	994	993	Eq. CH_2 rocking
1028	1028	1030	Eq. C–C str.
1052	1045	1046	Eq. C–C str.
1076	1073	1075	Eq CH_2 twisting
1129	1115	1115	Eq CH_2 twisting
1138			Ax. CH_2 twisting
	1175	1177	Eq. CH_2 twisting
1184	1192	1192	Eq. CH_2 wagging
1209			Ax. CH_2 wagging
1225			Ax. CH_2 wagging
1264	1263	1265	Eq. CH_2 wagging
1302	1293		Ax. CH_2 scissoring
1343	1334	1334	Eq. CH_2 scissoring
	1384		Ax. CH_2 scissoring
1446	1444	1444	Eq. CH_2 scissoring
2570			ν(SH)
2853	2850	2851	Eq. CH_2 str.
2892	2888	2886	Eq. CH_2 str.
2938	2931	2930	Eq. CH_2 str.

[a] Based on [17,20].
[b] Skew-boat C–S stretching discussed in [16].
[c] Fermi resonance with the C–S stretching band at ~720 cm^{-1} and the combination band at ~709 (=280 + 429) cm^{-1}.

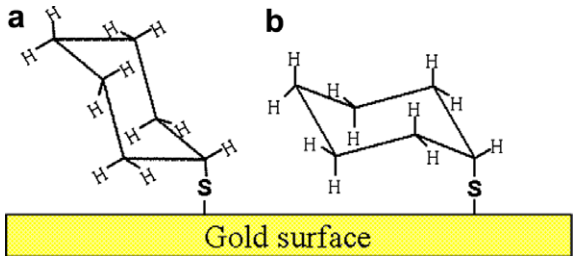

Fig. 2. Structural formula of two energetically stable conformers of CHT adsorbed on a gold surface. (a) Equatorial conformer and (b) axial conformer.

Table 2
Stabilities of the equatorial and axial forms of chair and boat conformers of CHT[a,b]

Energy difference[c]	Chair form	Boat form
Equatorial	0.0	29.3
Axial	5.7	33.7

[a] Unit: kJ/mol.
[b] Basis set with B3LYP/6-31G++(d,p).
[c] Energy difference based on the most stable equatorial chair form of CHT.

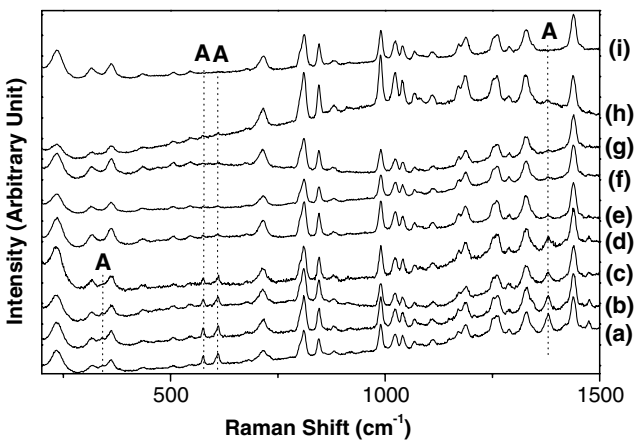

Fig. 3. SERS spectra in the region between 200 and 1500 cm^{-1} taken for CHT in aqueous gold nanoparticles at different concentrations of (a) 1×10^{-5}, (b) 2×10^{-5}, (c) 3×10^{-5}, (d) 4×10^{-5}, (e) 5×10^{-5}, (f) 6×10^{-5}, (g) 7×10^{-5}, (h) 8×10^{-5}, and (i) 9×10^{-5} M. "*A*" designates the vibrational bands characteristic of the axial conformer.

monolayers of CHT are more favorably formed on the gold surface at high concentrations. Fig. 4 depicts the relative peak height intensities of the axial band of the CH_2 scissoring bands at ~1380 cm^{-1} with respect to the corresponding equatorial band at ~1440 cm^{-1}, measured in the concentration range of 1×10^{-5}–9×10^{-5} M. It appears that both the equatorial and axial conformers coexist on Au at low concentrations, but the equatorial conformer could dominate at high concentrations.

Assuming the average diameter of the gold particles to be ~35 nm from the TEM images, the surface area of a single particle is calculated to be ~3850 nm^2. A single particle is composed of ~1.8×10^6 Au atoms and the concentration

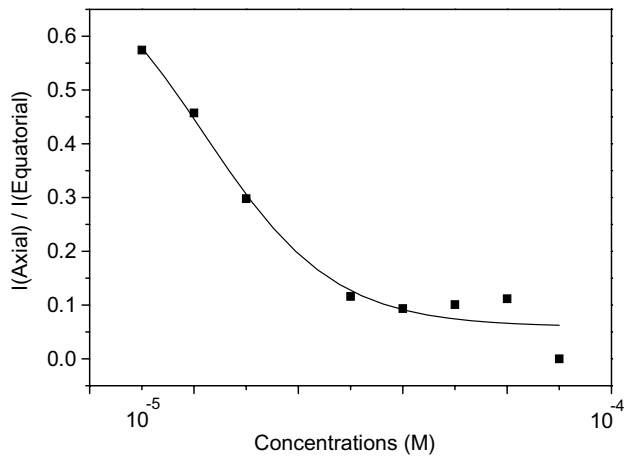

Fig. 4. Relative SERS intensities of the axial band of CH_2-scissoring bands at ~1380 cm^{-1} with respect to the corresponding equatorial band at ~1440 cm^{-1}, measured in the concentration range of 1×10^{-5}–9×10^{-5} M.

of the Au nanoparticles is estimated to be approximately 7.2×10^{-10} M. A concentration of 1.3×10^{-5} M is required for full monolayer coverage of the surface, and it is expected that 1×10^{-5} M should be near the full coverage limit. It was found that the vibrational modes that could be ascribed to the axial chair conformer were weaker than those of the equatorial conformer. This difference in vibrational modes is thought to exist because the equatorial conformation is more protruded than the axial one, as previously concluded by the interpretation of STM images [18].

The axial conformer is expected to be energetically unfavorable due to strong steric repulsions. Nonetheless, it is not absolutely certain whether a concentration-dependent adsorbate structure is associated entirely with thermodynamics or is due to the kinetics of the self-assembly of CHT on gold. Also, only an average orientation of the species could be obtained with respect to the several crystal planes that exist on the Au nanoparticle surfaces, the most prevalent of which is the (1 1 1) facet [9]. It is also possible that CHT may have a slightly different orientation on each facet.

To understand molecular-scale features of the CHT monolayer adsorbed on a gold surface, STM imaging was performed on CHT SAM samples on a Au(1 1 1) surface formed after immersion for one day in a 1 mM solution at room temperature. It is expected that these typical SAM preparation conditions are sufficient to obtain a fully covered SAM, as is the case for many organosulfur SAMs [21]. On the other hand, it was realized that CHT have a different conformation depending on the type of gold substrates. Therefore, it is hard to compare the STM results obtained from Au(1 1 1) surface and Raman results obtained from gold nanoparticles or gold plates. However, although the direct comparison of STM and Raman results is rather difficult, molecular-scale STM results can give us meaningful information about two-dimensional packing structure of energetically stable CHT SAMs on Au(1 1 1).

It could be helpful for understanding structural behavior of CHT adsorbed on gold surfaces. Moreover, we have to mention that the concentrations (1×10^{-5}–9×10^{-5} M) for the SERS experiments are much lower than that for the STM measurements (1 mM). For the monolayer coverage of the Au nanoparticle surfaces, the concentration of $\sim 10^{-5}$ M is sufficient. At a high concentration of 1 mM, the SERS spectra on Au nanoparticles are found to be interfered by unexpected strong ethanol solvent signals. In addition, we could easily obtain thermodynamically stable CHT SAMs on Au(1 1 1) in a 1 mM CHT solution for STM imaging. However, we found that it is hard to observe molecular-scale STM image of CHT SAMs on Au(1 1 1) formed in a CHT solution of $\sim 10^{-5}$ M, which may be due to a lower structural stability depending on surface coverage. STM images in Fig. 5 show that the adsorption of CHT molecules on a Au(1 1 1) surface results in well-ordered two-dimensional (2D) SAMs (region A) containing vacancy islands (region B), as is the case of conventional alkanethiols [21,22]. This result is rather surprising because it is expected that CHT would not form ordered SAMs due to a high-flexible cyclic ring. We believe that the formation of ordered SAMs is driven mainly by an energy gain resulting from the formation of S–Au bonds during self-assembly.

Interestingly, we often observed a few bright molecular spots in the well-ordered domains, as indicated by the white arrows in Fig. 5a. These molecular features were not observed in single-component alkanethiol or aromatic SAMs. Therefore, the bright spots can be ascribed to dynamic conformational changes of CHT in SAMs containing two conformers, as suggested by SERS measurements in Fig. 3 or Fig. 6. We also observed one molecular vacancy in the 2D monolayer, as indicated by the black arrow. The molecularly-resolved STM image in Fig. 5b shows that CHT SAMs are composed of two different molecular spots in the unit cell, suggesting the existence of two different adsorption configurations, i.e., equatorial and axial chair conformers. This STM result is consistent with the SERS result showing the existence of two conformers in the monolayer. We assume that the bright large and small molecular spots correspond to the equatorial and axial chair conformers, respectively. Additional structural details were described in another recent communication [18].

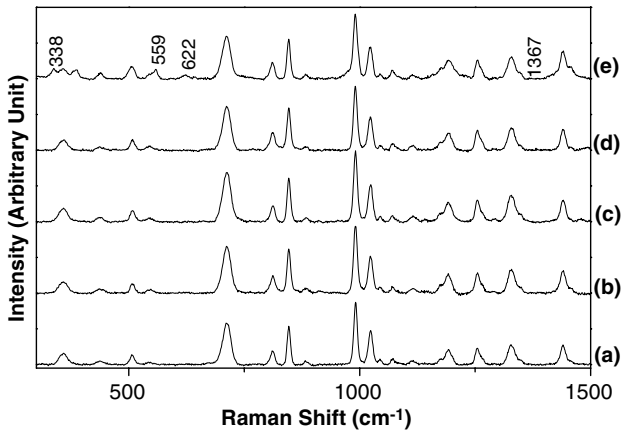

Fig. 6. SERS spectra in the region between 300 and 1450 cm^{-1} of CHT on a roughened gold plate at different temperatures of (a) −40, (b) 0, (c) 10, (d) 20, and (e) 40 °C. The frequencies of the most important diagnostic bands for the axial conformer at 338, 559, 622, and 1367 cm^{-1} for 40 °C are placed in the figure.

From the fundamental viewpoint, it is very important to understand the conformational dynamics of CHT adsorbed on metal surfaces as a function of surface temperature. We have tried to examine the axial/equatorial conformational changes of CHT molecules on SERS-active Au or Ag substrates by temperature-dependent Raman spectroscopy. However, it is hard to obtain good quality SERS spectra at higher temperatures or on Au substrates. At surface temperatures above room temperature, both adsorbates as well as metal atoms can diffuse rapidly on the surfaces to stabilize surface energy [21]. As a result, obtaining good quality SERS spectra for CHT on metal

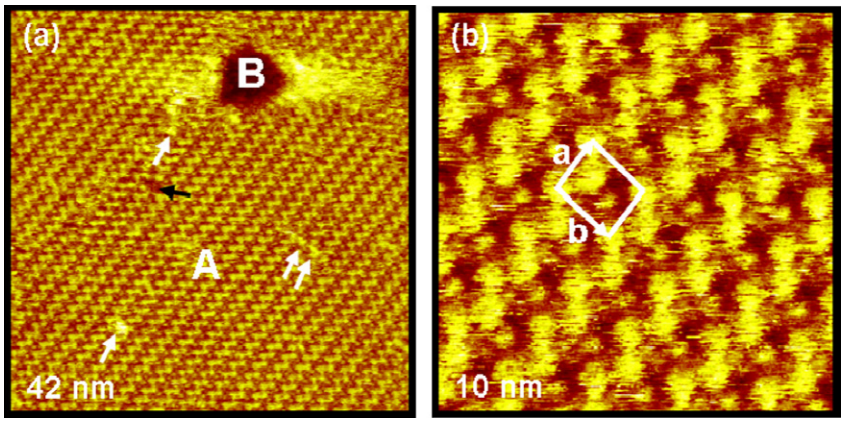

Fig. 5. STM images showing the formation of well-ordered SAMs on Au(1 1 1) surface by the adsorption of CHT molecules on Au(1 1 1). (a) A 42 nm × 42 nm STM image showing several bright molecular spots (white arrows) and one molecular vacancy (black arrow) in the ordered domain. (b) A 10 nm × 10 nm STM image showing two bright molecular features in the unit cell, suggesting the existence of equatorial and axial conformers in CHT SAMs.

surfaces at high temperatures seems unlikely. Fig. 6 shows the axial/equatorial conformational changes of CHT molecules adsorbed on gold plate with varying surface temperature. The spectral features on an Au plate become increasingly complicated with changing temperature. Among several changes, it is noteworthy that the peaks corresponding to the axial conformer at 580 and 614 cm^{-1} almost disappeared below room temperature. This result strongly implies that the ring conversion of axial to equatorial conformers can easily occur at low temperature to form energetically more favorable conformers, and is therefore a thermodynamically controlled process. The mole fraction of the axial form is assumed to decrease by half in lowering the temperature from 40 °C to −40 °C. Significant spectral changes occur by lowering temperatures. Complex features at 338, 559, 622, and 1367 cm^{-1} which could be ascribed to the axial conformer were found to disappear at a low temperature of 20 °C to −40 °C, as also observed in the case of high concentrations of CHT. Our results also support the notion that the equatorial conformer is only slightly more stable than the axial conformer. In our experiments, SERS spectra changed quite readily with the heating and cooling of the sample, indicating that the energy barrier of the axial/equatorial chair conformers should be quite low.

4. Conclusions

The effect of bulk concentration and surface temperature on the conformational change of cyclohexanethiol (CHT) adsorbed on gold nanoparticle and plate surfaces was examined using surface-enhanced Raman scattering (SERS). It was found that CHT molecules exist on Au surfaces in the form of both equatorial and axial conformers. The relative proportion of axial and equatorial conformers in the monolayer strongly depends on the bulk concentration. Both the equatorial and axial chair conformers of CHT were found to exist at low bulk concentrations near the monolayer coverage limit, whereas, the equatorial chair conformer appeared to be dominant at high bulk concentrations. High-resolution STM images revealed the presence of two stable conformers in the monolayers as well as the existence of meta-stable conformers, which may be due to dynamic ring conversion. In addition, the peaks assigned to the axial conformer almost disappeared below room temperature, suggesting that the ring conversion of the axial conformer can easily take place to form energetically more favorable equatorial conformers at low temperatures. This process can therefore be considered to be a thermodynamically controlled process.

Acknowledgements

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